

Propositional Logic

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Propositions

- ▶ Propositional logic is a branch of mathematics that studies the logical relationships between propositions (or statements, sentences, assertions) taken as a whole, and connected via logical connectives
- ▶ Proposition is a declarative statement

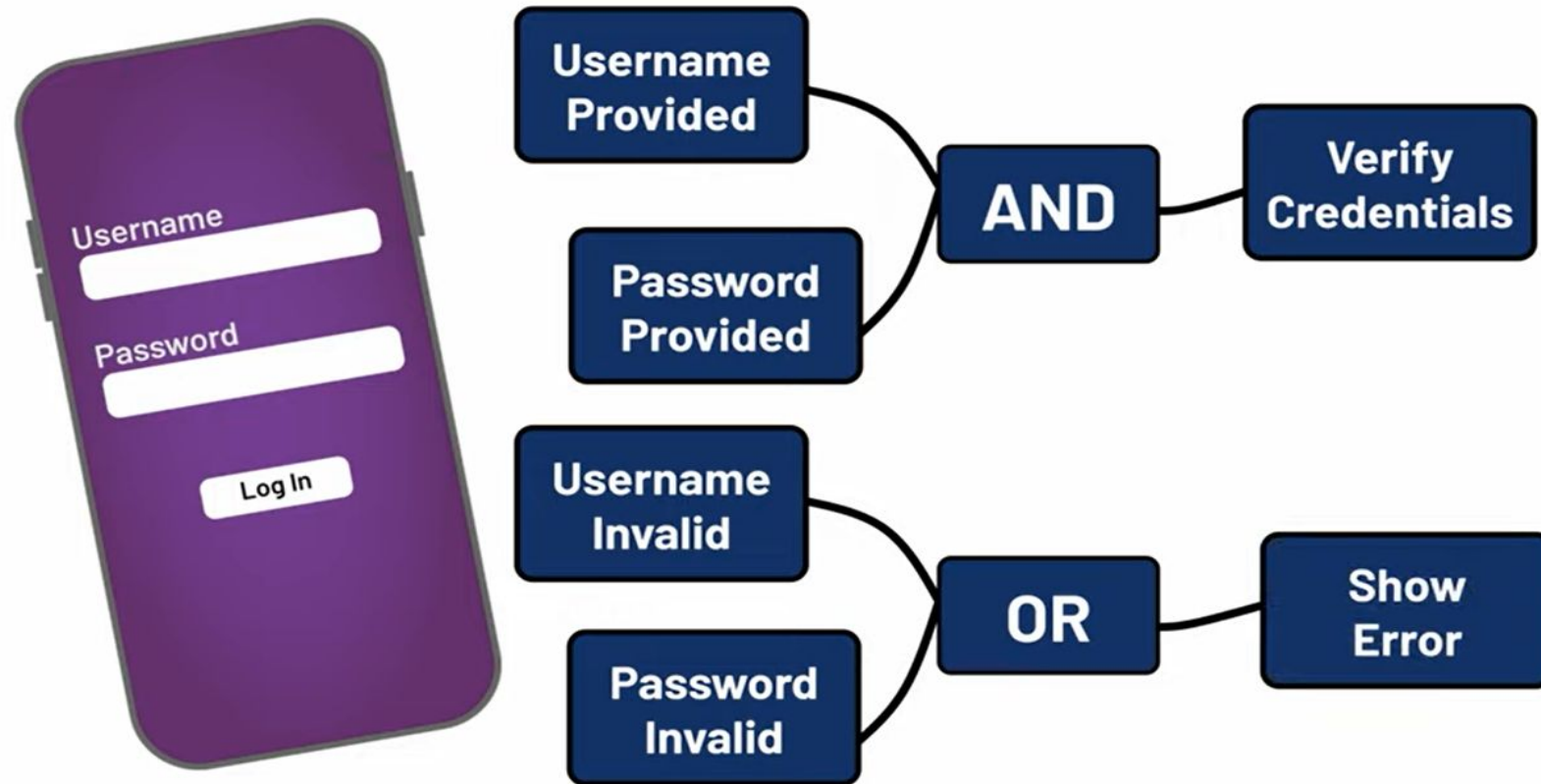
- ▶ Logical connectives:

- ▶ AND
- ▶ OR
- ▶ NOT
- ▶ IF-THEN

Statement	Truth Value
The sun rises in the East and sets in the West.	True
$1 + 1 = 2$	True
'b' is a vowel.	False

- ▶ A statement s is True/False where $s \in S$ has a definite truth value.

Propositions



Non-Propositions

- A statement s is non-propositional where $s \in S$ has a indefinite truth value.
 - ▶ “What time is it?” (Question)
 - ▶ “Go out and play.” (Command)
 - ▶ “ $x + 1 = 2$ ” (Open sentence — depends on the value of x)

Logical Connectives

Symbol	Words	Term	Example
	AND	Conjunction	
	OR	Disjunction	
	XOR	Exclusive OR	
	Implies	Implication	
	If and only If	Biconditional	
	Not	Negation	

Truth Table of Propositional Logic

➤ Negation:

- ▶ If p is a proposition, then the negation of p is denoted by $\neg p$, which when translated to simple English means- "*It is not the case that p* " or simply "*not p* ". The truth value of $\neg p$ is the opposite of the truth value of p .

T	F
F	T

Truth Table of Propositional Logic

➤ Conjunction:

- ▶ For any two propositions p and q , their conjunction is denoted by $p \wedge q$, which means "p and q". The conjunction $p \wedge q$ is True when both p and q are True, otherwise False.

T	T	T
T	F	F
F	T	F
F	F	F

Truth Table of Propositional Logic

➤ Disjunction:

- ▶ For any two propositions p and q , their disjunction is denoted by $p \vee q$, which means "p or q". The disjunction $p \vee q$ is True when either p or q is True, otherwise False.

T	T	T
T	F	T
F	T	T
F	F	F

Truth Table of Propositional Logic

➤ Exclusive Or:

- ▶ For any two propositions p and q , their exclusive or is denoted by $p \oplus q$, which means "either p or q but not both". The exclusive or $p \oplus q$ is True when either p or q is True, and False when both are true or both are false.

T	T	F
T	F	T
F	T	T
F	F	F

Truth Table of Propositional Logic

► Implication:

- For any two propositions p and q , the statement "if p then q " is called an implication and it is denoted by $p \rightarrow q$. In the implication $p \rightarrow q$, p is called the hypothesis or antecedent or premise (a thing that existed before or logically precedes another) and q is called the conclusion or consequence. The implication is $p \rightarrow q$ is also called a conditional statement. The implication is false when p is true and q is false otherwise it is true.
- "If it is Friday, then it is raining"
- "If student is good, then he/she gets A+"

T	T	T
T	F	F
F	T	T
F	F	T

Truth Table of Propositional Logic

➤ Implication:

- ▶ Implication guarantees that when p and q are true then the implication is true. But the implication does not guarantee anything when the premise p is false. There is no way of knowing whether or not the implication is false since p did not happen. This situation is similar to the "Innocent until proven Guilty" stance, which means that the implication $p \rightarrow q$ is considered true until proven false. Since we cannot call the implication $p \rightarrow q$ false when p is false, our only alternative is to call it true.
- ▶ This follows from the Explosion Principle which says: "A False statement implies anything" Conditional statements play a very important role in mathematical reasoning, thus a variety of terminology is used to express $p \rightarrow q$.
- ▶ "If p , then q ", " p is sufficient for q ", " q when p ", "a necessary condition for p is q ", " p only if q ", " q unless $\neg p$ ", " q follows from p ".
- ▶ Example, "If it is Friday then it is raining today" is a proposition which is of the form $p \rightarrow q$. The above proposition is true if it is not Friday (premise is false) or if it is Friday and it is raining, and it is false when it is Friday but it is not raining.

Truth Table of Propositional Logic

➤ Biconditional or Double Implication:

- ▶ For any two propositions p and q , the statement "p if and only if(iff) q" is called a biconditional and it is denoted by $p \leftrightarrow q$. The statement $p \leftrightarrow q$ is also called a bi-implication. $p \leftrightarrow q$ has the same truth value as $(p \rightarrow q) \wedge (q \rightarrow p)$. The implication is true when p and q have same truth values, and is false otherwise.
- ▶ "p is necessary and sufficient for q", "if p then q, and conversely", "p if q"
- ▶ "It is raining today if and only if it is Friday today."

T	T	T
T	F	F
F	T	F
F	F	T

Truth Table of Propositional Logic

- ▶ P: Good mobile phones are not cheap.
- ▶ Q: Cheap mobile phones are not good.
- ▶ **Consider the following statements:**
 - ▶ L: $P \rightarrow Q$
 - ▶ M: $Q \rightarrow P$
 - ▶ N: P is equivalent to Q
- ▶ Find, N is True or False.